

Shirley Zhang

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EDUCATION

Stanford University

Master of Science in Computer Science

Concentration: Information Management and Analytics

GPA: 3.90, Anticipated Graduation: December 2026

- **Relevant Courses:** Decision Making Under Uncertainty (Reinforcement Learning), Machine Learning (Python), Introduction to Big Data Systems (SQL), Operating Systems (C++), Mining Massive Datasets

Northwestern University

Bachelor of Science in Computer Science

Murphy Scholar, Summa Cum Laude

GPA: 3.99, Graduated June 2025

- **Relevant Courses:** Scalable Software Architecture (AWS), Introduction to Networking, Foundations of Security, Game Design and Development (C#), Foundations of Data Science, Design and Analysis of Algorithms

SKILLS

Programming Languages: Python, SQL, C++, JavaScript, Java, C#, HTML, CSS

Tools and Frameworks: React.js, MySQL, Pandas, PySpark, Airflow, Numpy, Tableau, MongoDB, AWS, Scikit-learn, TensorFlow, Matplotlib, Unity, Arduino

EXPERIENCE

Data Analyst Intern (ML & Data Engineering Focus)

March 2025 – June 2025

Tesla

- Designed and deployed a production ML pipeline processing fleet-scale time-series vehicle telemetry to predict brake system degradation, with model outputs integrated into live monitoring and diagnostics tools.
- Built an end-to-end data pipeline with a human-in-the-loop workflow to surface high-risk vehicles, accelerating thermos-related issue detection and root-cause investigation.
- Drove cross-functional debugging of production data and firmware regressions, partnering with firmware and engineering teams to identify systemic issues and improve reliability of brake system diagnostics.

Software Engineering Intern

June 2025 – August 2025

PwC (PricewaterhouseCoopers)

- Configured a multi-agent Langflow system wrapped around an internal RAG pipeline to route queries between retrieval and conversational agents, with support for translation and response post-processing.
- Contributed to internal Retrieval-Augmented Generation (RAG) prototypes by implementing backend evaluation logic and assisting with data retrieval workflows to improve document relevance and response quality.
- Refactored full-stack data storage and frontend rendering logic to address bugs caused by hierarchical structuring and cascading UI behavior, improving overall platform stability.

Teaching Assistant

September 2023 – December 2023

Northwestern University

- Lead biweekly exercises and discussions, simplifying core concepts such as recursion and the ethical usage of technology.
- Managed a cohort of 10 students in class, providing personalized support to help them apply basic programming principles.

PROJECTS

Differ: a Platform for Experiential Computing

December 2023 – September 2025

Delta Lab, Northwestern University

- First author on a CHI 2026 submission, presenting Differ as a platform for reasoning about contextual differences in human experience design and supporting case studies with real-world scenarios.
- Co-led research on computational models that enable system designers to evaluate differences in machine representations of abstract human experiences across different reference systems.
- Designed and implemented a lightweight data analysis framework to connect human concept expression with machine interpretation, surfacing actionable insights for inclusive and context-aware design.

Data Analysis Platform Integration

September 2023 – March 2025

NUSolar Software Team Lead, Northwestern University

- Led the integration of a data analysis system that parses vehicle CAN data, automating data storage and processing on AWS.
- Led the implementation of two separate visualization pipelines: one for the driver dashboard to display real-time data, and another for backend time-series analysis to support engineering diagnostics.
- Utilized SQL for efficient data querying and Python for processing, ensuring that the platform operates across distributed systems.